ACCV #802

Author Response: TriPoseFusion: Robust Multi-View 3D Driver Pose Fusion with Outlier-Aware Canonicalization

To all reviewers. Prompted by R-LAYQ's observation that PA-MPJPE exceeds MPJPE in Tab. 3 (corrected results explicitly requested), we audited our full evaluation stack. It confirmed the reviewers' concerns and found four defects: (i) a transposed rotation and norm-ratio scale in the Procrustes alignment; (ii) a zero-gradient gate-entropy regularizer (clamp after log); (iii) a degenerate shoulder fallback collapsing whole frames; (iv) uncalibrated constructed extrinsics in the triangulation reference, causing per-session metric-scale errors (median 2.4×). All are fixed (regression tests included); the reference is rebuilt by per-sequence self-calibration with anthropometric anchoring (reprojection 9.60 → 3.31 px; shoulder-width std 0.369 → 0.015 m). All numbers below use the corrected code and answer the reviewers' explicit requests; fixes, calibration, and tests will be released.

Reviewer LAYQ

W1 — metric anomaly. Cause: defects (i)+(iii). On synthetic rotated/scaled data the old code returns PA error 0.7–2.4 m where the truth is 0.04; the fix recovers the noise level. Corrected, same masks (a same-mask unaligned MPJPE is now also reported): canonicalized mean fusion 0.971/0.564 m on the original reference — the reversal is gone; 0.186/0.060 m on the calibrated one.

W2 — fallback rule. Your inference is exactly right; the missing rule: temporally nearest valid shoulder axis within the clip, else a default lateral axis, with a parallel-case guard. Pseudocode added to §3.3; model- and eval-side implementations unified.

W3 — independent reference. In-cabin mocap being infeasible, we instead validate zero-shot against *Drive&Act's official reference* (R-xS2o A2); all conclusions hold on both the original and the calibrated reference.

W4 — narrow the learned-component claims. We do: corrected, the full model ≡ mean fusion (0.1857/0.0593 vs. 0.1858/0.0595, fold-0 val), so the claims are narrowed to *canonicalization* + *uniform fusion* as the working core; the learned components' value is corruption robustness, not clean accuracy (R-xS2o A1); the 0.412 m magnitude: R-xS2o A2. Efficiency: 0.87M params, 3.3 MB, 6.2 ms/frame CPU.

Minor. Revision adds the exact 52-joint mapping table, frame-rate-normalized derivatives, GroupKFold splits/durations, and per-view imbalance analysis; broader joint-limit/biomechanical evaluation: acknowledged as future work.

Reviewer 19jf

W1 — reference reliability under occlusion. Now quantified per joint: valid-rate head/shoulder–neck 1.00, hands 0.34; hands dominate the corrected error (0.316 m vs. torso 0.067, shoulder–neck 0.033). The revision reports hands separately, adds a torso-only protocol, and states the pipeline precisely: manual checks apply to the 2D inputs; triangulation is automatic DLT with a 40 px filter, now per-sequence self-calibrated, with per-sequence uncertainty statistics. External validation: R-xS2o A2.

W2 — canonicalization mechanism. The gain is the removal of per-view depth/scale inconsistency, not the fixed rig: the same procedure transfers zero-shot to Drive&Act's different camera geometry, where canonicalized fusion (23.1 mm PA) again beats every single view (24.0–31.9 mm).

Minor — module necessity. Agreed — see R-LAYQ W4 and R-xS2o A1.

Reviewer xS2o

A1 — "rectify the neural collapse" (W1–W2). The collapse is real but not forced by the entropy regularizer — defect (ii) made its gradient exactly zero. Diagnosis: retraining with the fix, with λ_{gate}=0, and with a leave-one-view-out teacher all still yield 0.3333 gates — the collapse is architectural (under a median pseudo-teacher, uniform fusion is a trivial optimum); claims narrowed accordingly (R-LAYQ W4); W2 likewise confirmed — TCN/velocity add no clean-input accuracy. Rectification: corruption-augmented training restores functional gating — a 1.9K-param head assigns weight 0.037 to a zeroed view, degrading only +1–5% (mean fusion: +18%; weight exactly 0 for missing views); an 8K residual head improves clean MPJPE −14% (0.128 vs. 0.149, 5-fold).

A2 — Drive&Act evaluation (request 2, "mandatory"). Done, zero-shot (official reference, different rig, no training on it): canonicalize+mean **23.1 mm** PA-MPJPE, full model **23.7 mm** (rigid-MPJPE 31.6/32.5 mm) — zero-shot, against the *fully supervised* 22.6–30.4 mm SOTA you cite; label-free per-rig adaptation of the 8K head reaches **18.6 mm** (−23%). The paper's 0.412 m magnitude was the defect-(iv) scale artifact, not pose error (W3); PCK@0.05/0.02 added (M3).

A3 — deep-learning baselines (request 3). UPose3D, Learnable Triangulation, and Faster VoxelPose consume images or calibrated volumes and are inapplicable to our keypoint-only interface, which by design avoids image-level fusion. On the identical input we add the applicable learned family: confidence-weighted LT-style regression (1.9K), set-attention with uncertainty head (28K; Spearman 0.72), residual head (8K; best), our original architecture (870K; worst) — 16 configurations, 5-fold.

Minor. SAM3D architecture/domain details added (M1); reference circularity (M2): R-19jf W1 and A2.